

add time
dimension

study

+ play

SYSTEMS SIMULATION

Systems Analysis & Design

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validation

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UNIVERSIDAD DISTRITAL
FRANCISCO JOSÉ DE CALDAS

Outline

1 Basic Concepts /

2 Cellular Automata /

3 Digital Twins /



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- 1 Basic Concepts
- 2 Cellular Automata
- 3 Digital Twins



What is a simulation?

- **Simulations** are a type of real-world representation, but rarely exactly like the real world.
- Sometimes you need to **test** or **experiment** with expensive use cases. Simulations let you **play** with different inputs, configurations, hyperparameter optimizations.
- Also, there are dangerous or hard-to-reach **scenarios** where **simulations** become the best option.
- You should define the **level of detail** well enough to represent the expected behavior, without failing due to either high complexity or lazy simplicity.

high complexity



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- chemical processes
- construction



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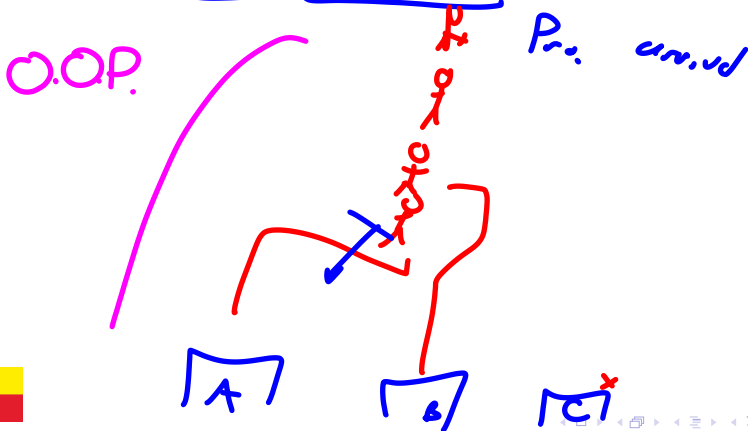
Events and Stochastic Processes

- It is typical to play with **event probabilities**, creating **stochastic behaviors**.
- One way to **simulate many systems** is to use **event-based models**.
Embrace the chaos and Murphy's Law.

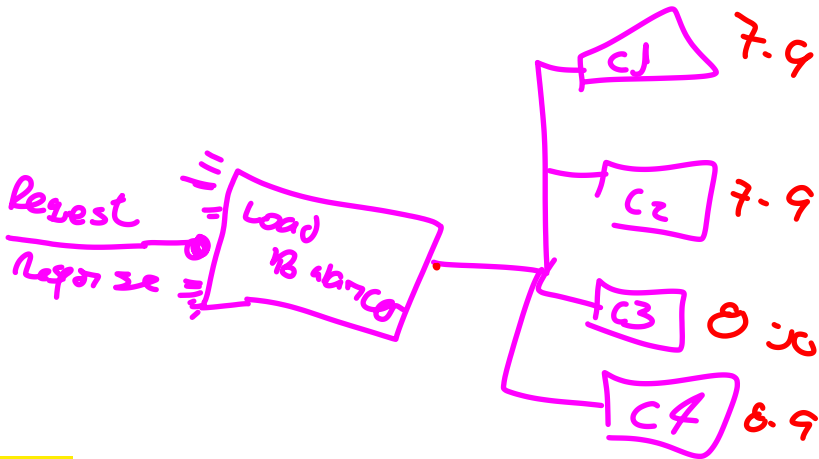


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Case Study: Load Balancer in a Cloud Architecture



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Cellular Automata

state changes

- Cellular Automata are a discrete model defined by a grid of cells, each with a state.
- The state of a cell is updated based on the states of its neighbors.

cells

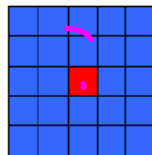
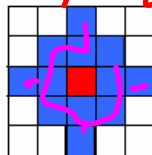
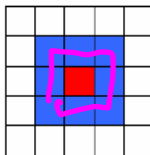
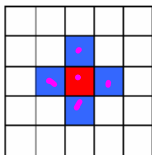
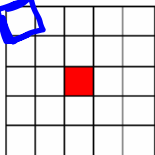


Figure: Cellular Automata Typical Neighborhoods



continuous

time

list

object

video games

flow



Game of Life

- **Game of Life** is a **cellular automaton** devised by the British mathematician **John Horton Conway** in **1970**.
- It is a **zero-player game**, meaning that its **evolution** is determined by its **initial state**, requiring no further input.
- Rules:

chaos

$t = 0$

- Any live cell with fewer than two live neighbors dies, as if by underpopulation.
- Any live cell with two or three live neighbors lives on to the next generation.
- Any live cell with more than three live neighbors dies, as if by overpopulation.
- Any dead cell with exactly three live neighbors becomes a live cell, as if by reproduction.



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- It is a **zero-player game**, meaning that its **evolution** is determined by its **initial state**, requiring no further input.
- Rules: *→ deterministic*
 - Any **live** cell with **fewer than two** live neighbors **dies**, as if by underpopulation. *0 - dead, 1 - live*
 - Any **live** cell with **two or three** live neighbors **lives** on to the next generation.
 - Any **live** cell with **more than three** live neighbors **dies**, as if by overpopulation.
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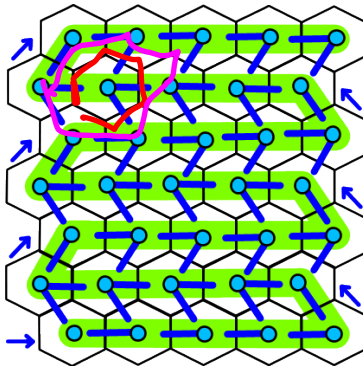


for $i \in \{ \dots \}$
 for $j \in \{ \dots \}$
 $sum = \sum_{i-1, j-1}^{i+1, j+1} M_{ij}$



HoneyComb Cellular Automata

- **HoneyComb Cellular Automata** is a different topology where a cell has six neighbors.
- This representation has different dispersion properties, which can be more interesting.



→ sparse matrix



SIR Model

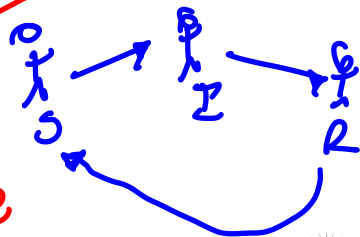
→ SARS (2004)

→ discrete

- **SIR Model** is a compartmental model used to represent the transmission of a contagious disease.
- The **model** divides the population into three compartments: **S** for the number of susceptible individuals, **I** for the number of infected individuals, and **R** for the number of recovered individuals.
- The model is defined by the following differential equations where β is the transmission rate and γ is the recovery rate:

$$\begin{aligned} \frac{dS}{dt} &= -\beta \cdot S \cdot I \\ \frac{dI}{dt} &= \beta \cdot S \cdot I - \gamma \cdot I \\ \frac{dR}{dt} &= \gamma \cdot I \end{aligned}$$

Prob.



↓
~ Game Of Life

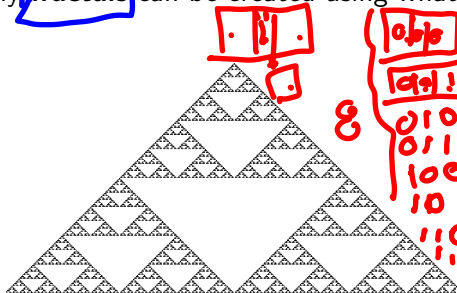


Chaotic Systems

- **Chaotic Systems** are a class of dynamical systems that exhibit sensitive dependence on initial conditions. *8 bits $\rightarrow$ 1 byte*
- This means that the future behavior of the system is highly dependent on the initial conditions.
- The **Lorenz System** is a well-known example of a chaotic system.
- Using **cellular automata** to simulate chaotic systems is a common practice. Many **fractals** can be created using what are called chaotic rules.

Handwritten binary calculations:
 $00011000 = 24$
 $01080000 = 10$

Handwritten notes:
 4D $\sim$ 3.7D
 3D 2.1D
 2D 1.9D



Handwritten label: Sierpinski

Handwritten binary calculations:
 $0101 = 5$
 $1011 = 11$
 $0110 = 6$
 $1001 = 9$
 $1101 = 13$
 $1110 = 14$
 $1010 = 10$
 $0100 = 4$
 $1100 = 12$
 $0011 = 3$
 $1000 = 8$
 $0111 = 7$
 $1111 = 15$



Turing Morphogenesis

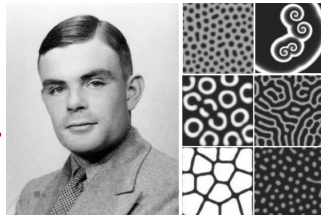
— FSM
— A.I.

— Cryptography
— ALife → Math + Biology

- **Turing Morphogenesis** is a theory of biological development that explains how patterns form in living organisms. *→ shapes/skin*
- The theory is based on the idea that chemical signals can interact to create patterns in living organisms. *↓ ↓ ↓*
- The reaction-diffusion model is a common way to simulate **Turing morphogenesis**. *↔ d.f. reaction / d.f. eq. disc.*
- The model is defined by a set of reaction and diffusion equations that describe how the chemical signals interact.

cells $< r < d$

Alan Turing



— 1953



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What are Digital Twins?

very realistic simulation

- A **Digital Twin** is a virtual model that accurately reflects a physical object or system.
- **Digital twins** enable real time monitoring, simulation, and analysis of the physical counterpart.
- They are widely used in manufacturing, smart cities, healthcare, and energy systems.
- By integrating sensor data, **digital twins** help in predictive maintenance and process optimization.



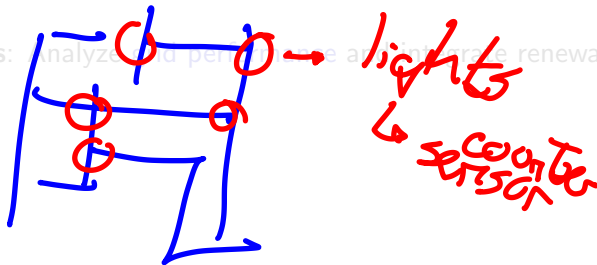
Applications of Digital Twins

- **Manufacturing:** Monitor equipment performance and predict failures.
- **Urban Planning:** Model city infrastructure to optimize traffic and resource use.
- **Healthcare:** *Train* Simulate patient-specific treatments and optimize procedures.
- **Energy Systems:** *People* Analyze grid performance and integrate renewable energy sources.



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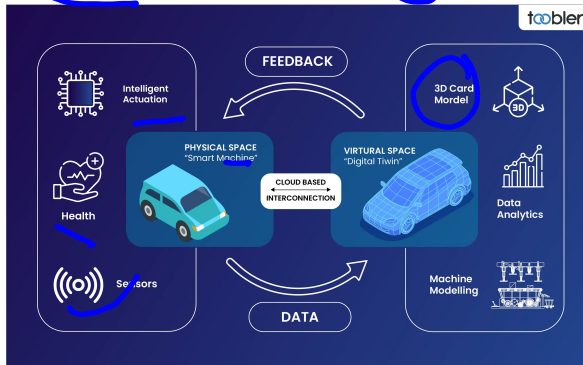
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Benefits of Digital Twins

- Enhanced **decision-making** through **real-time data analysis**.
- Improved efficiency and reduced operational costs.
- ~~Better understanding of system behavior and performance.~~
- Increased collaboration between teams and stakeholders.



Challenges of Digital Twins

- Data integration and management.
- Ensuring data security and privacy.
- Maintaining the accuracy of the digital twin.
- Managing the complexity of the physical system.
- Ensuring interoperability between different systems and platforms.
- Scalability for common large-scale systems.
- Cost of implementation and maintenance.
- User training and adoption.
- Regulatory compliance and standardization.
- Ethical considerations in data usage and analysis.
- Environmental impact of digital twin technologies.



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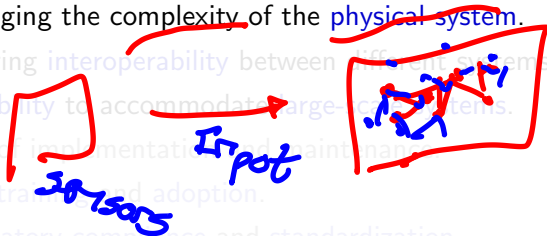
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Inputs



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• D.T. → economy

health → public

Tesla ~ > 100 sensors



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new versions → cloud



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ChatGPT → nice



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cloud
sensors
Internet
of things



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Thanks!

Questions?



Repo: <https://github.com/EngAndres/ud-public/tree/main/courses/systems-analysis>

